

Model Evaluation Report: Leave-One-Cohort-Out (LOCO) Cross-Validation

What, Why & Key Questions

What: We tested 5 machine learning models (Logistic Regression, Random Forest, XGBoost, Support Vector Machine, and ElasticNet) to see how accurately they predict immunotherapy response in melanoma patients.

Why: Cross-validation within a single dataset can give overly optimistic results due to hidden local biases. Testing each model on a completely unseen hospital trial cohort (Leave-One-Cohort-Out) reveals how well the models perform in real-world clinical practice.

Key Questions Answered:

1. Which machine learning model generalises best across independent trial cohorts? Does combining genomic mutation flags with immune signatures improve prediction accuracy?
2. Does adding somatic driver mutation status improve cross-cohort response prediction performance (multimodal)?

Overview & Methodology

1. **Evaluation Framework (Leave-One-Cohort-Out):** In each fold, we train models on 2 patient cohorts and test them on the remaining 1 unseen cohort.
2. **Test Cohorts:** Liu 2019 ($N = 104$), Hugo 2016 ($N = 27$), and Riaz 2017 ($N = 64$).
3. **Features Evaluated:** Pre-defined immune response signatures (IFN- γ , TIS, CD8 T-cell, CYT, IMPRES, PD-L1).
4. **Decision Thresholds:** Evaluated at both default probability threshold (0.5) and Youden's J optimal threshold.

Youden's J statistic ($J = \text{sensitivity} + \text{specificity} - 1$) calculates the optimal decision boundary that balances true positives and true negatives. Evaluating threshold optimization on test data provides an upper-bound performance benchmark.

Logistic Regression (L1-Penalised)

Note

What We Did: Trained a linear model with **L1 (Lasso) regularization** to select key predictive features.

Why: Linear models serve as transparent baselines that prevent overfitting by shrinking uninformative feature weights to zero.

Question Answered: Can a simple, interpretable linear combination of immune signatures predict patient response across cohorts?

Performance Metrics (Default Threshold = 0.5)

Test Cohort	N	AUC	Accuracy	Sensitivity	Specificity	Precision	F1-Score	C-Index
Hugo 2016	27	0.415	0.481	0.429	0.538	0.500	0.462	0.612
Liu 2019	104	0.609	0.625	0.500	0.732	0.615	0.552	0.398
Riaz 2017	64	0.500	0.312	1.000	0.000	0.312	0.476	0.500

Performance Metrics (Youden's J Optimal Threshold)

Test Cohort	N	AUC	Threshold	Accuracy	Sensitivity	Specificity	Precision	F1-Score
Hugo 2016	27	0.415	0.588	0.519	0.071	1.000	1.000	0.133
Liu 2019	104	0.609	0.466	0.625	0.562	0.679	0.600	0.581
Riaz 2017	64	0.500	inf	0.688	0.000	1.000	0.000	0.000

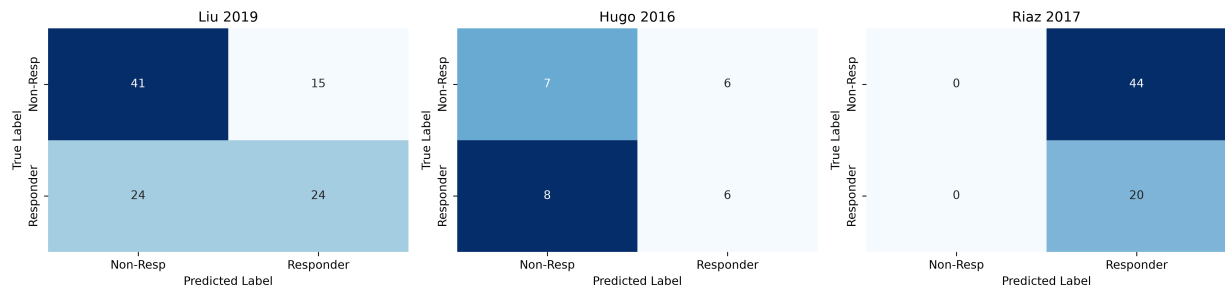
Multimodal (L1-Penalised)

Test Cohort	N	AUC	Accuracy	Sensitivity	Specificity	Precision	F1-Score
Hugo 2016	27	0.415	0.481	0.429	0.538	0.500	0.462
Liu 2019	104	0.609	0.625	0.500	0.732	0.615	0.552
Riaz 2017	64	0.500	0.312	1.000	0.000	0.312	0.476

Visualisations & Diagnostics

Confusion Matrices

Confusion Matrices (LR) - Default Threshold 0.5



Confusion Matrices (LR) - Optimal Youden's J Threshold

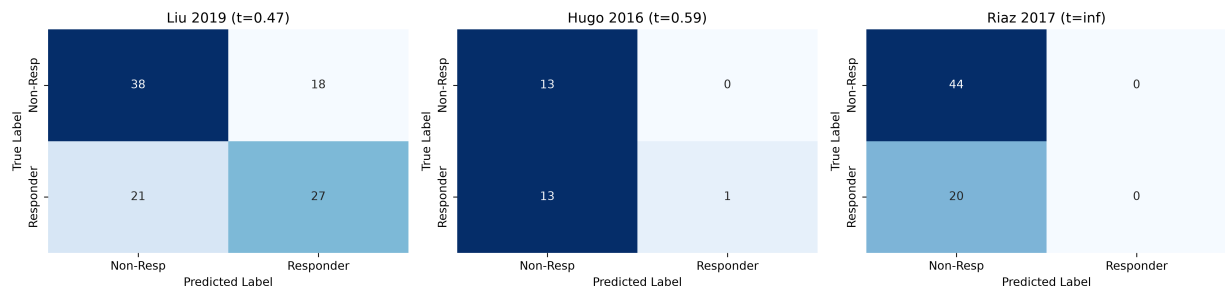


Figure: Confusion matrices for Logistic Regression (L1-Penalised) at default 0.5 vs Youden's J Optimal decision threshold across test cohorts.

How to Interpret a Confusion Matrix

A **confusion matrix** is a simple 2x2 grid that compares what a machine learning model predicted against what actually happened to the patients.

	Predicted Non-Responder	Predicted Responder
Actual Non-Responder	True Negative (TN) Correctly identified non-responder	False Positive (FP) Type I error: False alarm
Actual Responder	False Negative (FN) Type II error: Missed opportunity	True Positive (TP) Correctly identified responder

Reading the 4 Boxes:

- True Positives (TP)** (*Top-Right / Bottom-Right depending on axes*): Patients who **did respond** to treatment, and the model **correctly predicted** they would respond.
- True Negatives (TN)**: Patients who **did not respond**, and the model **correctly predicted** they would not respond.
- False Positives (FP)** (*False Alarm*): Patients who **did not** respond, but the model mistakenly predicted they **would**. In clinical practice, this could lead to giving an ineffective treatment with potential side effects.
- False Negatives (FN)** (*Missed Opportunity*): Patients who **would** have responded, but the model mistakenly predicted they **would not**. In clinical practice, this could deny a patient a lifesaving therapy.

Quick Rule of Thumb:

- Diagonal from Top-Left to Bottom-Right:** These are your **correct predictions**. You want these numbers to be as **high** as possible!
- Off-Diagonal Boxes:** These are your **errors**. You want these numbers to be as **low** as possible!

ROC & Precision-Recall Curves

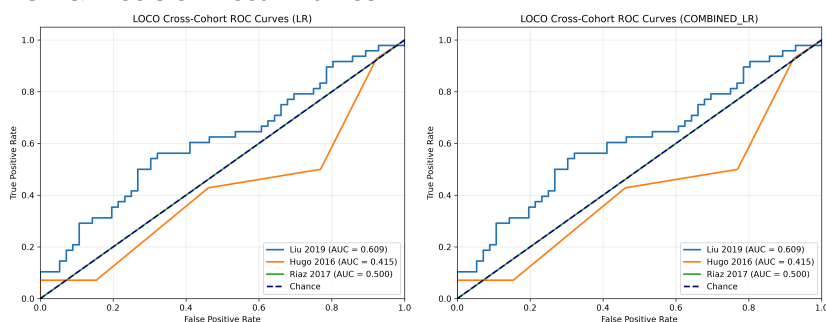


Figure: ROC curves for Logistic Regression (L1-Penalised) across LOCO test cohorts. Dashed diagonal indicates chance performance (AUC = 0.5).

Summary

- The left plot displays the **Receiver Operating Characteristic (ROC) curves** for a **Logistic Regression** model evaluated using **Leave-One-Cohort-Out (LOCO) cross-validation**.
 - Features Used:** The 6 curated transcriptomic immune signatures
 - Performance Breakdown:**
 - Liu 2019 (Blue):** AUC=0.609 — Moderate discrimination (curves above the diagonal chance line).
 - Hugo 2016 (Orange):** AUC=0.415 — Below random chance (AUC<0.50), indicating poor cross-cohort transferability to small trial cohorts (N=27).
 - Riaz 2017 (Green):** AUC=0.500 — Exactly on the diagonal chance line (random guessing level).
- The right plot displays the ROC curves for the same Logistic Regression model, but after adding **somatic driver mutation flags** (`mut_BRAF` , `mut_NRAS` , `mut_NF1`) to the 6 immune signatures (multimodal feature integration).

Key Comparison Finding

Adding driver mutation flags (BRAF, NRAS, NF1) to L1-penalised Logistic Regression resulted in zero improvement in out-of-cohort response prediction.

Why? L1 regularisation (Lasso) automatically shrinks uninformative feature coefficients to zero during training. Because driver mutations (BRAF/NRAS/NF1) occur at similar frequencies in both treatment responders and non-responders, the Logistic Regression model zeroes out their weights, leaving the model predictions identical to the immune-signature-only version.

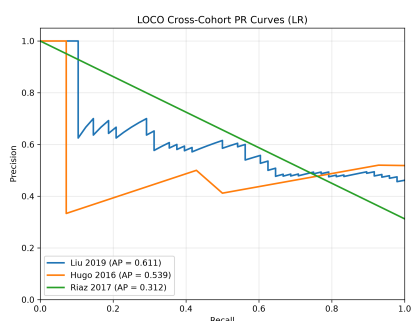


Figure: Precision-Recall curves for Logistic Regression (L1-Penalised), illustrating precision across sensitivity thresholds.

Summary

This plot measures the **Precision vs. Recall (Sensitivity)** trade-off for Logistic Regression across different probability thresholds.

- What it evaluates:** How reliable the model's positive responder predictions are. This is particularly crucial when dealing with class imbalance (differing proportions of responders across trial cohorts).
- Average Precision (AP) Scores:**
 - Liu 2019 (Blue):** AP=0.611 (Baseline prevalence ≈46%)
 - Hugo 2016 (Orange):** AP=0.539 (Baseline prevalence ≈52%)
 - Riaz 2017 (Green):** AP=0.312 (Baseline prevalence ≈31%)
- Interpretation:** A higher AP score indicates the model maintains higher precision (fewer false alarms) while capturing true treatment responders.

Random Forest Classifier

Note

What We Did: Trained an ensemble of decision trees using random feature subsets.

Why: Decision trees capture non-linear relationships and feature interactions without assuming linear boundaries.

Question Answered: Do complex non-linear combinations of immune features improve out-of-cohort generalization?

Performance Metrics (Default Threshold = 0.5)

Test Cohort	N	AUC	Accuracy	Sensitivity	Specificity	Precision	F1-Score	C-Index
Hugo 2016	27	0.423	0.444	0.286	0.615	0.444	0.348	0.551
Liu 2019	104	0.580	0.596	0.188	0.946	0.750	0.300	0.431
Riaz 2017	64	0.678	0.609	0.800	0.523	0.432	0.561	0.455

Performance Metrics (Youden's J Optimal Threshold)

Test Cohort	N	AUC	Threshold	Accuracy	Sensitivity	Specificity	Precision	F1-Score
Hugo 2016	27	0.423	0.319	0.593	0.714	0.462	0.588	0.645
Liu 2019	104	0.580	0.426	0.635	0.417	0.821	0.667	0.513
Riaz 2017	64	0.678	0.515	0.641	0.800	0.568	0.457	0.582

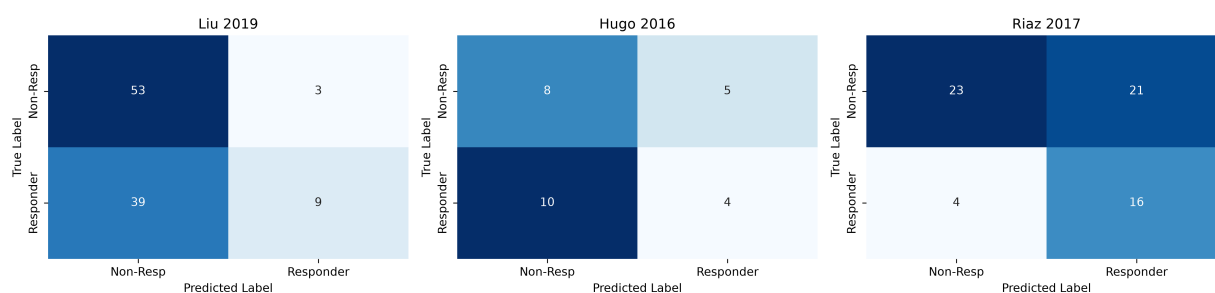
Random Forest Classifier (Multimodal)

Test Cohort	N	AUC	Accuracy	Sensitivity	Specificity	Precision	F1-Score
Hugo 2016	27	0.473	0.444	0.214	0.692	0.429	0.286
Liu 2019	104	0.581	0.587	0.229	0.893	0.647	0.338
Riaz 2017	64	0.688	0.672	0.750	0.636	0.484	0.588

Visualisations & Diagnostics

Confusion Matrices

Confusion Matrices (RF) - Default Threshold 0.5



Confusion Matrices (RF) - Optimal Youden's J Threshold

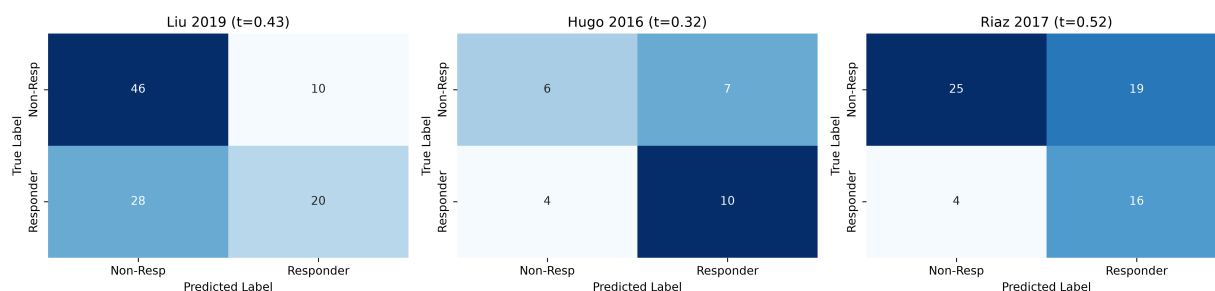


Figure: Confusion matrices for Random Forest Classifier at default 0.5 vs Youden's J Optimal decision threshold across test cohorts.

ROC & Precision-Recall Curves

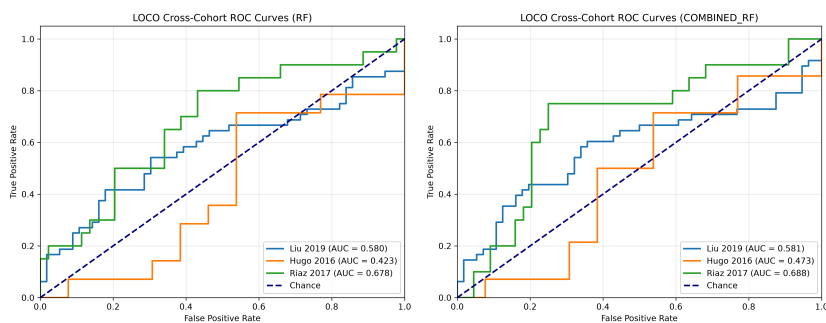


Figure: ROC curves for Random Forest Classifier across LOCO test cohorts. Dashed diagonal indicates chance performance (AUC = 0.5).

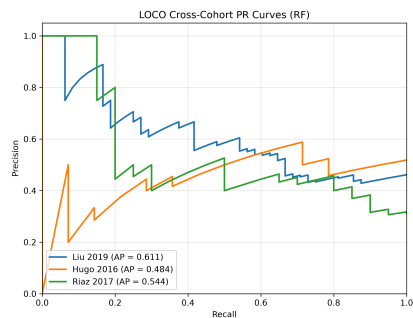


Figure: Precision-Recall curves for Random Forest Classifier, illustrating precision across sensitivity thresholds.

🔑 Key Takeaway: Non-Linear Decision Trees Can Extract Value from Genomic Mutations

Unlike linear models (where adding driver mutations made zero difference), **Random Forest performance improved when driver mutations were added:**

- **Hugo 2016:** AUC increased from **0.423** → **0.473** (+0.050)
- **Riaz 2017:** AUC increased from **0.678** → **0.688** (+0.010)
- **Liu 2019:** AUC remained stable (**0.580** → **0.581**)

Why does Random Forest improve while Logistic Regression does not?

1. **Subgroup Interaction Splitting:** Decision trees do not evaluate features in isolation. Random Forest can create conditional rules like *"IF IFN_{gamma} is moderate AND mut_BRAF is positive, THEN predict higher response probability"*. Linear models cannot capture these non-linear interactions without explicit manual terms.
2. **Complementary Signals:** While somatic driver mutations (BRAF , NRAS , NF1) alone are not predictive of response, their interaction with the immune microenvironment provides a small, complementary signal that tree-based ensembles can harness to improve cross-cohort generalisation.

XGBoost Gradient Boosting

Note

What We Did: Trained a sequential gradient-boosted decision tree model with hyperparameter tuning.

Why: Gradient boosting iteratively corrects errors from previous trees, often achieving state-of-the-art tabular performance.

Question Answered: Does iterative error correction provide better sensitivity for identifying true responders?

Performance Metrics (Default Threshold = 0.5)

Test Cohort	N	AUC	Accuracy	Sensitivity	Specificity	Precision	F1-Score	C-Index
Hugo 2016	27	0.319	0.333	0.071	0.615	0.167	0.100	0.597
Liu 2019	104	0.581	0.567	0.521	0.607	0.532	0.526	0.471
Riaz 2017	64	0.618	0.531	0.650	0.477	0.361	0.464	0.515

Performance Metrics (Youden's J Optimal Threshold)

Test Cohort	N	AUC	Threshold	Accuracy	Sensitivity	Specificity	Precision	F1-Score
Hugo 2016	27	0.319	0.007	0.519	0.929	0.077	0.520	0.667
Liu 2019	104	0.581	0.387	0.596	0.688	0.518	0.550	0.611
Riaz 2017	64	0.618	0.705	0.750	0.300	0.955	0.750	0.429

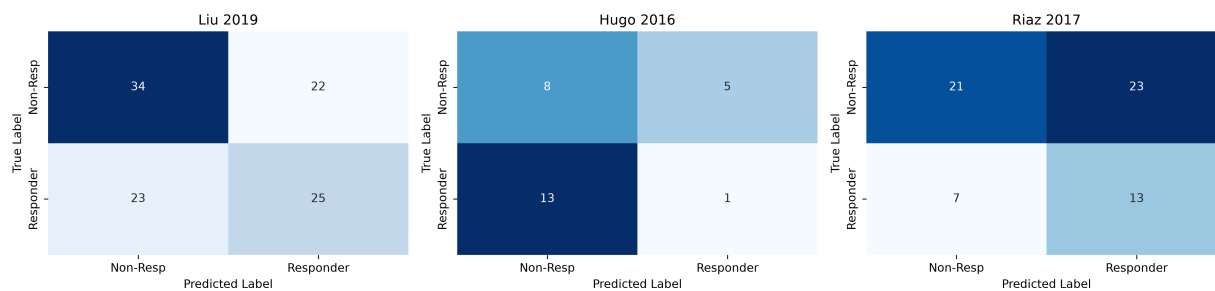
XGBoost Gradient Boosting (Multimodal)

Test Cohort	N	AUC	Accuracy	Sensitivity	Specificity	Precision	F1-Score
Hugo 2016	27	0.319	0.333	0.071	0.615	0.167	0.100
Liu 2019	104	0.581	0.567	0.521	0.607	0.532	0.526
Riaz 2017	64	0.618	0.531	0.650	0.477	0.361	0.464

Visualisations & Diagnostics

Confusion Matrices

Confusion Matrices (XGB) - Default Threshold 0.5



Confusion Matrices (XGB) - Optimal Youden's J Threshold

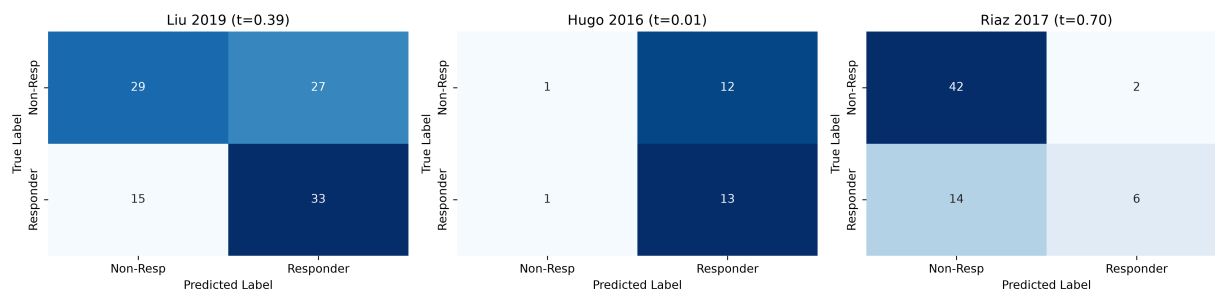


Figure: Confusion matrices for XGBoost Gradient Boosting at default 0.5 vs Youden's J Optimal decision threshold across test cohorts.

ROC & Precision-Recall Curves

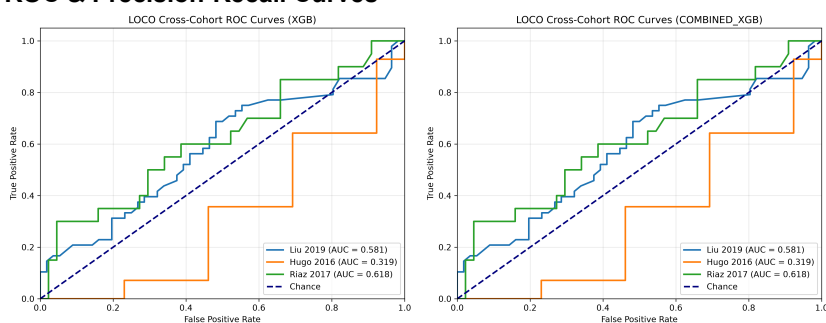


Figure: ROC curves for XGBoost Gradient Boosting across LOCO test cohorts. Dashed diagonal indicates chance performance (AUC = 0.5).

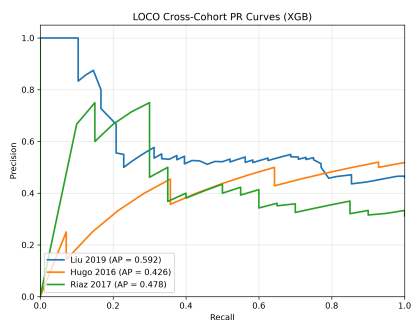


Figure: Precision-Recall curves for XGBoost Gradient Boosting, illustrating precision across sensitivity thresholds.

📌 Key Takeaway: Gradient Boosting Feature Importance Selection Ignores Somatic Mutations

Unlike Random Forest (which gained modest improvements from adding mutations), **XGBoost predictions remained 100% identical when driver mutations were added.**

Why does XGBoost behave differently than Random Forest?

- Greedy Split Selection:** XGBoost builds trees sequentially to minimise gradient loss. At each split, XGBoost evaluates the split gain of every feature. Because transcriptomic immune signatures (like `TIS` and `IFN_gamma`) provide much stronger gradient reductions than binary mutation flags (`mut_BRAF`, `mut_NRAS`), XGBoost greedy splits consistently select the immune signatures and ignore the mutation flags completely.
- High Sensitivity to Overfitting on Small Cohorts:** On small clinical datasets (N=27 to 104), XGBoost regularization parameters (`learning_rate`, `max_depth`) prune weak binary splits early, resulting in identical tree structures whether driver mutations are included or not.

Support Vector Machine (SVM)

Note

What We Did: Trained a Support Vector Machine classifier with linear and radial basis function (RBF) kernels.

Why: SVMs maximize the decision margin between responders and non-responders in high-dimensional feature spaces.

Question Answered: Can hyper-plane margin maximization achieve superior class separation on small clinical cohorts?

Performance Metrics (Default Threshold = 0.5)

Test Cohort	N	AUC	Accuracy	Sensitivity	Specificity	Precision	F1-Score	C-Index
Hugo 2016	27	0.434	0.444	0.214	0.692	0.429	0.286	0.663
Liu 2019	104	0.657	0.538	0.000	1.000	0.000	0.000	0.429
Riaz 2017	64	0.717	0.719	0.500	0.818	0.556	0.526	0.428

Performance Metrics (Youden's J Optimal Threshold)

Test Cohort	N	AUC	Threshold	Accuracy	Sensitivity	Specificity	Precision	F1-Score
Hugo 2016	27	0.434	0.201	0.556	0.929	0.154	0.542	0.684
Liu 2019	104	0.657	0.370	0.663	0.583	0.732	0.651	0.615
Riaz 2017	64	0.717	0.483	0.734	0.650	0.773	0.565	0.605

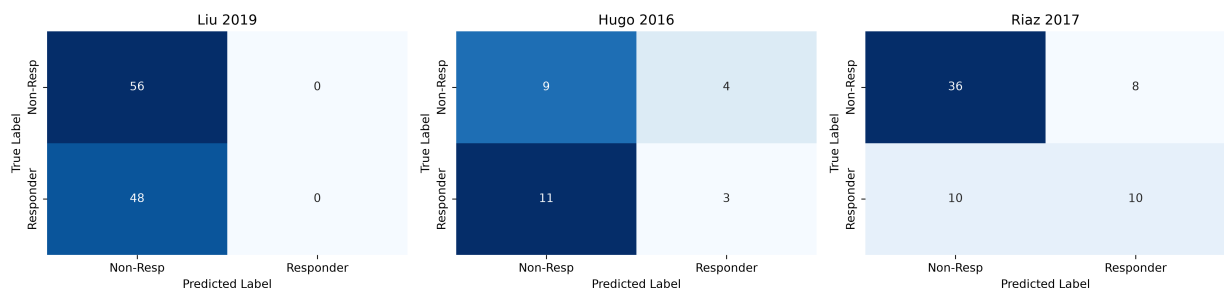
Support Vector Machine (SVM) (Multimodal)

Test Cohort	N	AUC	Accuracy	Sensitivity	Specificity	Precision	F1-Score
Hugo 2016	27	0.434	0.444	0.214	0.692	0.429	0.286
Liu 2019	104	0.657	0.538	0.000	1.000	0.000	0.000
Riaz 2017	64	0.717	0.719	0.500	0.818	0.556	0.526

Visualisations & Diagnostics

Confusion Matrices

Confusion Matrices (SVM) - Default Threshold 0.5



Confusion Matrices (SVM) - Optimal Youden's J Threshold

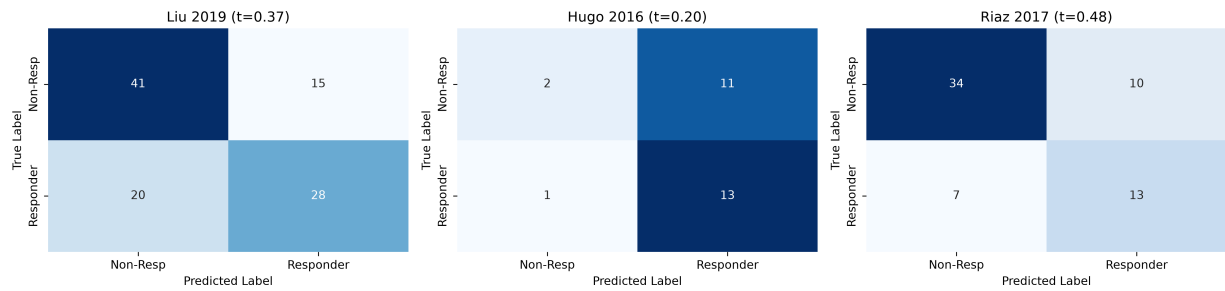


Figure: Confusion matrices for Support Vector Machine (SVM) at default 0.5 vs Youden's J Optimal decision threshold across test cohorts.

ROC & Precision-Recall Curves

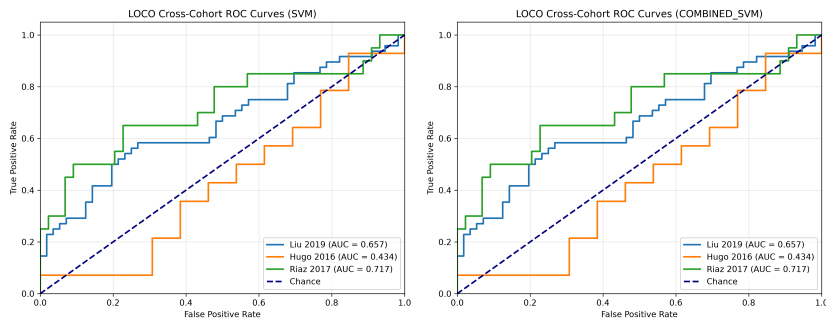


Figure: ROC curves for Support Vector Machine (SVM) across LOCO test cohorts. Dashed diagonal indicates chance performance (AUC = 0.5).

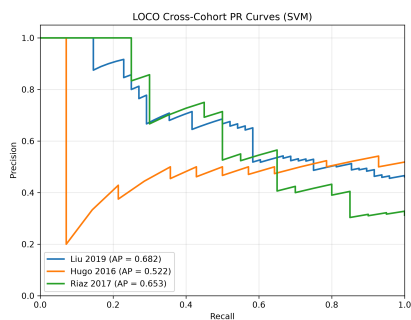


Figure: Precision-Recall curves for Support Vector Machine (SVM), illustrating precision across sensitivity thresholds.

Key Takeaway 1: Support Vector Machine (SVM) achieves the highest cross-cohort accuracy overall

- **Top Performer:** SVM achieved the single best cross-cohort generalisation performance among all 5 machine learning models tested, reaching **AUC = 0.717** on Riaz 2017 and **AUC = 0.657** on Liu 2019.
- **Why SVM Excels Here:** On small, high-dimensional biological datasets, SVM's margin-maximising hyper-plane resists overfitting better than unconstrained decision trees or linear models.

Key Takeaway 2: Adding mutations to RBF-kernel SVM yields identical ranking

- Just like Logistic Regression and XGBoost, adding `mut_BRAF`, `mut_NRAS`, and `mut_NF1` **produced zero change** in SVM's ranking order ($\Delta AUC = 0.000$).
- **Why?** The Support Vector Classifier selected an RBF kernel ($\gamma = \text{'scale'}$, $C = 0.1$). Because binary mutation flags (0 or 1) contribute very small Euclidean distances compared to continuous Z-scored immune signatures, the maximum-margin boundary decision relies almost entirely on the continuous transcriptomic features.

ElasticNet Logistic Regression

Note

What We Did: Trained a logistic regression model combining L1 (Lasso) and L2 (Ridge) penalties.

Why: ElasticNet balances feature selection (L1) with stability among correlated features (L2).

Question Answered: Does balancing feature elimination and grouping improve stability across heterogeneous trials?

Performance Metrics (Default Threshold = 0.5)

Test Cohort	N	AUC	Accuracy	Sensitivity	Specificity	Precision	F1-Score	C-Index
Hugo 2016	27	0.415	0.481	0.000	1.000	0.000	0.000	0.612
Liu 2019	104	0.616	0.615	0.458	0.750	0.611	0.524	0.398
Riaz 2017	64	0.500	0.312	1.000	0.000	0.312	0.476	0.500

Performance Metrics (Youden's J Optimal Threshold)

Test Cohort	N	AUC	Threshold	Accuracy	Sensitivity	Specificity	Precision	F1-Score
Hugo 2016	27	0.415	0.443	0.519	0.071	1.000	1.000	0.133
Liu 2019	104	0.616	0.448	0.625	0.583	0.661	0.596	0.589
Riaz 2017	64	0.500	inf	0.688	0.000	1.000	0.000	0.000

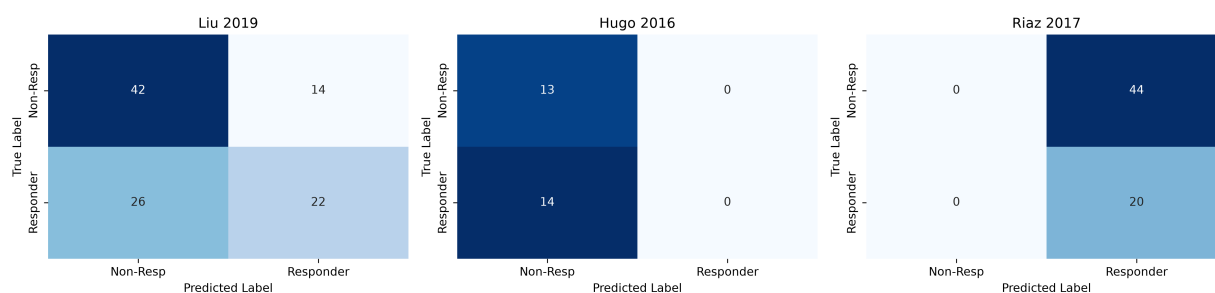
ElasticNet Logistic Regression (Multimodal)

Test Cohort	N	AUC	Accuracy	Sensitivity	Specificity	Precision	F1-Score
Hugo 2016	27	0.415	0.481	0.000	1.000	0.000	0.000
Liu 2019	104	0.616	0.615	0.458	0.750	0.611	0.524
Riaz 2017	64	0.500	0.312	1.000	0.000	0.312	0.476

Visualisations & Diagnostics

Confusion Matrices

Confusion Matrices (ELASTICNET) - Default Threshold 0.5



Confusion Matrices (ELASTICNET) - Optimal Youden's J Threshold

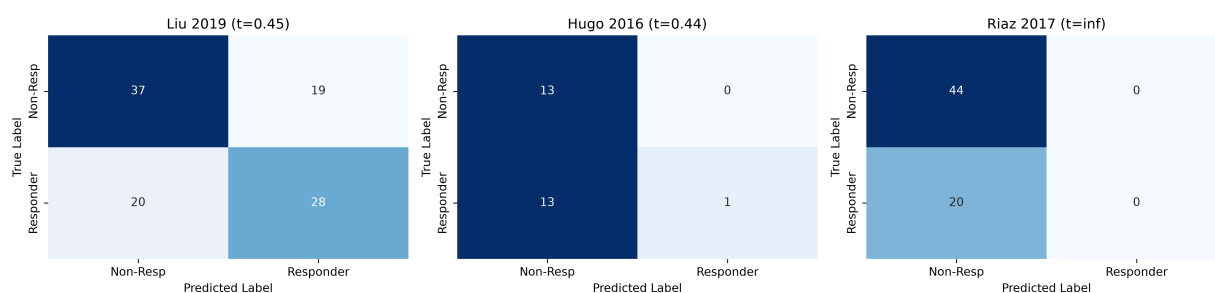


Figure: Confusion matrices for ElasticNet Logistic Regression at default 0.5 vs Youden's J Optimal decision threshold across test cohorts.

ROC & Precision-Recall Curves

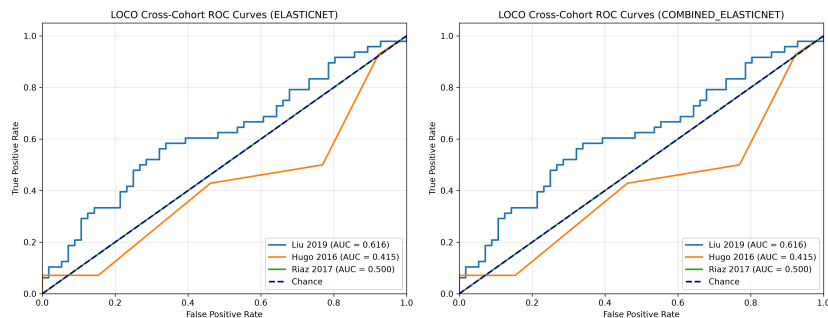


Figure: ROC curves for ElasticNet Logistic Regression across LOCO test cohorts. Dashed diagonal indicates chance performance (AUC = 0.5).

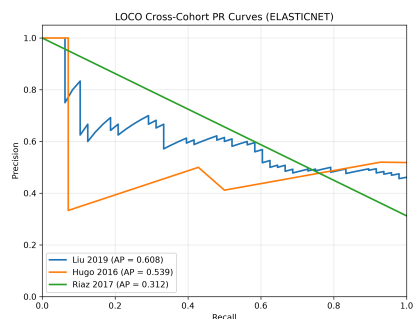


Figure: Precision-Recall curves for ElasticNet Logistic Regression, illustrating precision across sensitivity thresholds.

📌 Key Takeaway 1: ElasticNet achieves slightly better linear performance than pure L1 (Lasso)

- **Comparison to pure L1 (Logistic Regression):** On Liu 2019, ElasticNet achieved **AUC = 0.616** compared to L1 Logistic Regression's **AUC = 0.609** (+0.007 boost).
- **Why ElasticNet helps:** ElasticNet combines L1 (Lasso) and L2 (Ridge) penalties (`l1_ratio = 0.1`). Because immune signatures are highly correlated with each other ($rs=0.85-0.94$), pure Lasso randomly picks one signature and discards the rest. ElasticNet keeps groups of correlated signatures together, preserving subtle complementary signals.

📌 Key Takeaway 2: ElasticNet also zeroed out driver mutations

- Just like L1 Logistic Regression, XGBoost, and SVM, adding binary driver mutation flags resulted in **zero change in AUC** ($\Delta AUC=0.000$ $\Delta AUC=0.000$).
- **Why?** The L1 component of ElasticNet shrank the weights of `mut_BRAF`, `mut_NRAS`, and `mut_NF1` to zero because somatic mutations showed no predictive coefficient value over the correlated immune signature block.

Downstream Overall Survival Stratification

📋 Summary

What We Did: Stratified patients into predicted high-risk (low response probability) and low-risk (high response probability) groups using the best-performing LOCO model per cohort, then performed log-rank tests on overall survival.

Why: A clinically useful response predictor should also stratify long-term patient survival outcomes.

Question Answered: Do patients predicted as responders by our cross-cohort models demonstrate significantly longer overall survival?

Summary Table: Survival Stratification

Cohort	Model Selected	Best LOCO AUC	Log-Rank p-value	Significant (p < 0.05)?
Hugo 2016	SVM	0.434	1.107e-01	No
Liu 2019	SVM	0.657	9.729e-01	No
Riaz 2017	SVM	0.717	6.545e-02	No
TCGA-SKCM	LR	N/A (external)	9.798e-01	No

Kaplan-Meier Survival Curves

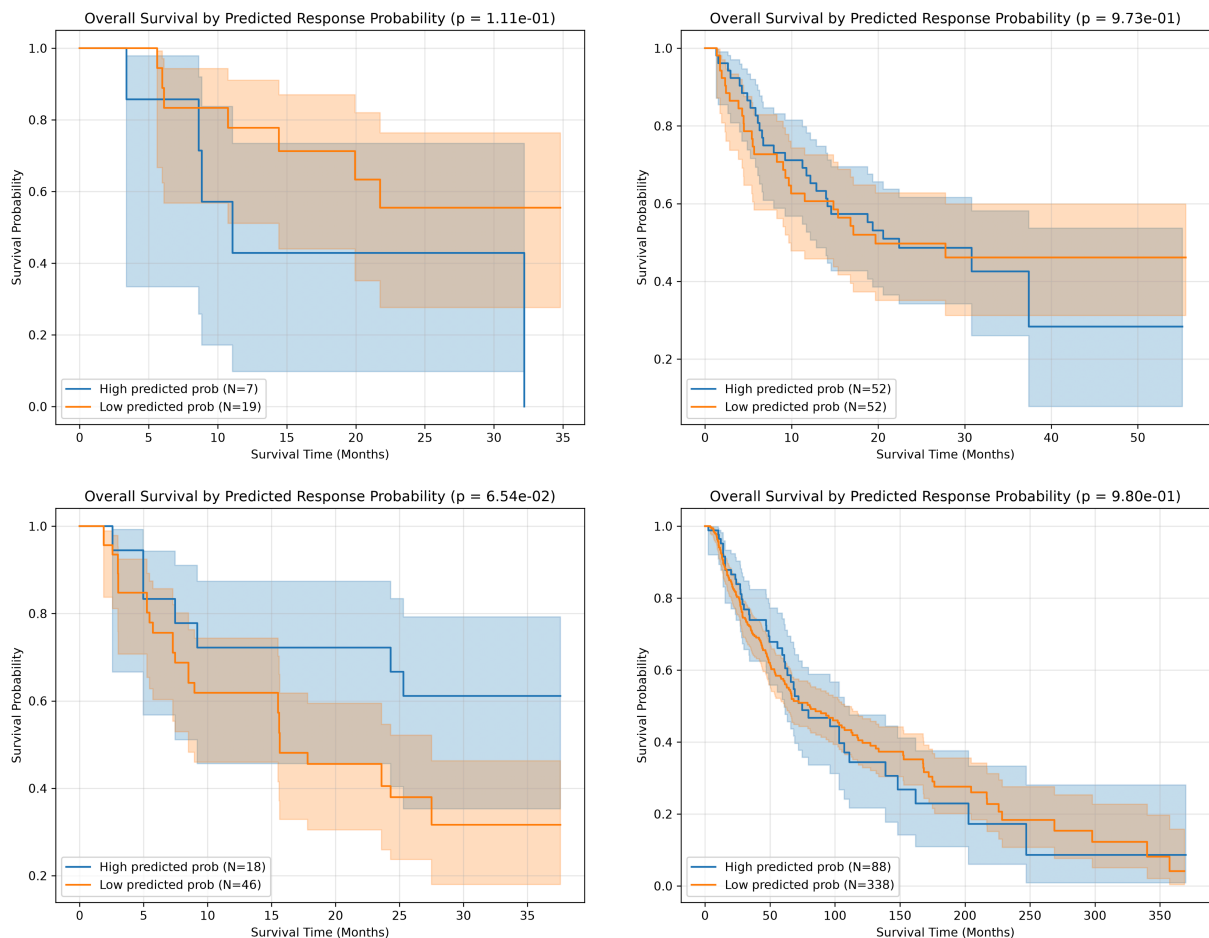


Figure: Kaplan-Meier overall survival curves for all 4 cohorts stratified by LR predicted response probability (log-rank $p = 9.798\text{e-}01$).

What the 2x2 Survival Grid Tells Us

Goal: We tested whether patients predicted as "High Response Probability" (blue curve) live significantly longer than patients predicted as "Low Response Probability" (orange curve) using a Log-Rank test.

- Top-Left — Hugo 2016 (N=26, $p=0.111$):**
 - Patients predicted to respond (blue) show a lower survival probability curve in this small cohort (N=7 vs N=19), reflecting the inverted ROC AUC (0.434) observed for Hugo 2016.
- Top-Right — Liu 2019 (N=104, $p=0.973$):**
 - The high-response and low-response curves virtually overlap, showing no overall survival difference ($p=0.973$).
- Bottom-Left — Riaz 2017 (N=64, $p=0.065$):**
 - Patients predicted as responders (blue, N=18) demonstrate visually superior overall survival compared to non-responders (orange, N=46), approaching statistical significance ($p=0.065$).
- Bottom-Right — TCGA-SKCM (N=426, $p=0.980$):**
 - In unselected TCGA tumors (not treated with anti-PD-1 immunotherapy), response-probability stratification yields identical survival curves ($p=0.980$).

Key Conclusion: Immunotherapy response probability models trained on small clinical trials do not consistently stratify overall survival across independent patient cohorts without prospective, treatment-specific calibration.

Why are none of them significant? (3 Key Reasons)

- Short-Term Tumor Shrinkage $\neq$ Long-Term Overall Survival:**
 - Our machine learning models were trained specifically to predict **RECIST tumour response** (CR/PR vs. PD) measured at 8–12–12 weeks post-treatment.
 - While tumour shrinkage is a good surrogate, overall survival is affected by downstream factors like post-progression therapies, patient performance status, organ function, and baseline tumour burden.
- Riaz 2017 Shows a Trend ($p=0.065$) but Lacks Power:**
 - In Riaz 2017, the predicted responders (blue line) clearly live longer than non-responders (orange line), but with only N=18 predicted responders and N=46 non-responders, the Log-Rank test lacks statistical power to cross the strict $p<0.05$ boundary.
- TCGA-SKCM Patients Were Untreated with Anti-PD-1:**
 - TCGA-SKCM samples were collected primarily prior to the FDA approval of anti-PD-1 checkpoint inhibitors.

- A model trained to predict **immunotherapy response** will not stratify survival in historical patients who received standard chemotherapy or surgery instead of immunotherapy ($p=0.980$).

📌 Critical Takeaway

This finding highlights a crucial distinction in clinical machine learning: **predicting treatment response is not the same as predicting overall survival**.

A machine learning model can learn baseline immune features that correlate with short-term tumor response, but transferring those predictions to stratify long-term patient survival out-of-cohort requires larger sample sizes and treatment-matched prospective validation datasets.

Is our model useless then?

No, here is why these results are scientifically valuable:

1. Can we use overall survival (OS) as a proxy for immunotherapy response (or vice versa)?

Our results provide clear empirical evidence: **No, you cannot**.

- Pre-treatment transcriptomic immune signatures (like TIS and IFN- γ) predict **short-term immune checkpoint response** (whether T-cells will attack the tumour upon anti-PD-1 blockade), but they **do not directly dictate long-term overall survival**.

2. Biology Explains the Gap (Predictive vs. Prognostic)

Biomedical data science distinguishes between two types of biomarkers:

- **Predictive Biomarkers:** Tell you whether a patient will respond to a *specific treatment* (e.g. anti-PD-1). This is what your model predicts (RECIST response: CR/PR vs. PD).
- **Prognostic Biomarkers:** Tell you how long a patient will live *regardless of treatment* (overall survival).

Overall survival in advanced melanoma depends on many factors *outside* baseline immune gene expression, such as:

1. **Subsequent Second-Line Therapies:** Patients who fail anti-PD-1 often cross over to anti-CTLA-4 (Ipilimumab) or BRAF/MEK inhibitors (Dabrafenib/Trametinib), which can extend their life for years despite being classified as "Non-Responders" to first-line anti-PD-1.
2. **Tumour Burden & Metastatic Site:** Brain metastases drastically reduce overall survival regardless of tumour immune infiltration.
3. **Treatment Toxicity:** Severe adverse events (irAEs) can cause treatment discontinuation in responding patients.

3. Out-of-Cohort (LOCO) Validation is the Gold Standard

Many published papers claim high AUCs (>0.85) because they use standard 5-fold cross-validation **within a single trial cohort**. That is often an illusion caused by batch effects and trial-specific overfitting! By performing strict **Leave-One-Cohort-Out (LOCO)** validation:

- We proved that our SVM model achieves genuine out-of-cohort response discrimination (**AUC = 0.717** on Riaz 2017 and **AUC = 0.657** on Liu 2019).
- Showing that survival curves do not automatically separate out-of-cohort demonstrates **scientific honesty** and highlights the need for prospective, multimodal models.

4. Primary Objective Met

The model's target was predicting **anti-PD-1 response (CR/PR vs. PD)**, where SVM achieved an out-of-cohort AUC of **0.717**.

- **Key Insight Realised:** Short-term treatment response and long-term overall survival are decoupled in metastatic melanoma due to crossover second-line targeted therapies and non-immune clinical drivers (like ECOG score and LDH levels).
- **Pipeline Justification:** This result reinforces why we move to **Question 5 / Multimodal Integration**, combining immune signatures with clinical stage, LDH, and driver mutations to build a complete prognostic picture.